

Patterns, Parity & Multiples — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (c) **8.** Each of the three gaps is + or -: $2 \times 2 \times 2 = 8$.

Q2 (c) **4n.** $4n = 2(2n)$ has 2 as a factor, so it is always even.

Q3 **24.** $n + (n+2) + (n+4) = 72 \Rightarrow n = 22$, middle = 24.

Q4 (b) **14.** $7 \times 29 = 203$ up to $7 \times 42 = 294 \Rightarrow 42 - 29 + 1$.

Q5 **True.** One of two consecutive numbers is always even.

Q6 (a) **Both true, R explains A.** Since $9 = 3 \times 3$, anything divisible by 9 is divisible by 3.

Short answer

Q7 **k-1, k-2, k-3, k-4, k-5, k-6.**

Q8 **Always even.** $2a + 2b = 2(a + b)$; 2 is a factor, so the value is a multiple of 2.

Q9 **36.** It must be divisible by $\text{LCM}(9, 4) = 36$.

Long answer

Q10 **7, 8, 9, 10.** $n + (n+1) + (n+2) + (n+3) = 4n + 6 = 34 \Rightarrow n = 7$.

Q11 (i) **Always even** — $4x + 6 = 2(2x + 3)$. (ii) **Sometimes even** — $3g + 5h$ has no common factor of 2; e.g. $g=h=1$ gives 8 (even) but $g=1, h=0$ gives 3 (odd).

Q12 **Always even.** $n + (n+1) + (n+2) + (n+3) = 4n + 6 = 2(2n + 3)$, which has 2 as a factor.

Total: 21 marks.